1. Introduction & Objectives

1.1 Introduction

This project involved designing and optimizing a Filtered PID Controller for a 3-Degree-of-Freedom (3DOF) helicopter system. The main challenge was getting the system to follow a reference elevation path which includes step inputs, steady-state cruise phases, and a parabolic landing ramp while also being able to handle a severe 2-second motor blackout at $t = 65s$. Balancing tracking performance with hardware safety during that kind of fault was the core engineering problem.

1.2 Project Objectives

My design goals focused on meeting five core engineering criteria to maximize performance and protect the hardware:

1. **Steady-State Error ($e_{ss} = 0$):** Eliminate any tracking offset during cruise by using a Type-1 structure with a pure integrator at the origin.

2. **Transient Response (5% Overshoot):** Keep bouncing between altitude steps within the 5% overshoot specification ($\zeta = 0.6901$).

3. **Disturbance Rejection (Recovery < 5s):** After the 2-second motor blackout at t = 65 s, bring the helicopter back to within ±0.01 rad of its cruise altitude in under 5 seconds.

4. **Sensor Noise Attenuation:** Use a derivative filter (τ = 0.012 s) to deal with quantizer noise from the 0.0015 rad sensor and prevent that from reaching the motors.

5. **Actuator Protection:** Minimize sudden voltage jumps ($\frac{du}{dt}$) to protect the motors and keep the control signal within the ±10 V plant limits.

2. Controller Selection and Specifications

I chose a **Filtered PID Controller** to stabilize the helicopter's elevation and track the reference path. A standard PID controller can suffer from sudden voltage spikes (noise chattering) because of the derivative term. Adding a low-pass filter to the derivative path fixes that and keeps the motors from being hit with sudden voltage jumps.

To get a fast response with minimal bouncing, i picked a design specification of 5% overshoot. Using the standard second-order formula, this gives a required damping ratio (ζ) of 0.6901. To stay safely away from actuator saturation, i chose a stable baseline natural frequency of ω_n= 1.1 rad/s.

3. Controller Design Method

I used the pole-placement matching coefficient method shown in class to calculate the exact PID gains. The given helicopter plant parameters are:

$a_1 = 0.09525$, $a_0 = 1.613$, $b_0 = 0.1675$

The idea is to get the closed-loop system to match a desired third-order characteristic equation,, where two poles are determined by our chosen ζ and ω_n, and a third real pole is placed at -α.

$$\Delta desired(s) = (s^2 + 2\zeta\omega_n s + \omega_n^2)(s + \alpha)$$

Matching coefficients gives the three gain formulas:

1. $Kd = \frac{(2\zeta\omega_n + \alpha - a1)}{b0}$
2. $Kp = \frac{(\omega_n^2 + 2\,\zeta\omega_n\alpha - a0)}{b0}$
3. $Ki = \frac{(\alpha\,\omega_n^2)}{b0}$

After testing different values in simulation (detailed in Section 5), the third pole was pushed out to α = 5.0 and the filter time constant tightened to τ = 0.012 s. This gave the integrator enough aggressiveness to recover quickly after the blackout without saturating the actuator. The final gains came out to:

Kd = 38.3461

Kp = 42.9140

Ki = 36.1194

In my MATLAB script, these were entered as:

numG = [38.3461, 42.9140, 36.1194];

denG = [0.012, 1, 0];

4. Closed-Loop Transfer Function

To analyze the overall system behavior, the closed-loop transfer function $T(s)$ is derived using the filtered PID controller $G_C(s)$ and the helicopter plant model $P(s)$ in a negative feedback loop:

$$T(s) = \frac{\left(G_c(s)P(s)\right)}{1 + G_c(s)P(s)}$$

Substituting the plant parameters and my final optimized controller gains (Kd = 38.3461, Kp = 42.9140, Ki = 36.1194) gives the fourth-order closed-loop transfer function:

$$T(s) = \frac{(6.4251s^2 + 7.2608s + 6.0500)}{0.012s^4 + 1.0011s^3 + 6.6433^2 + 8.8805s + 6.0500}$$

5. Tuning History and Simulation Iterations

Finding the best parameters required testing different settings in the simulation to lower the cost function value (J).

- **Trial 1** (ω_n = 2.0, α = 10.0): i started with an aggressive speed. This made the gains too high (Kp ≈179), causing the controller to hit the ±10 V voltage saturation block during the $t = 65\ s$ drop. This resulted in severe integrator windup, making the helicopter swing wildly and hit the ground, triggering a 1.50 crash penalty for a total cost of 105.31435.

- **Trial 2** (ω_n = 1.1, α = 3.3): i scaled back the speed to ω_n = 1.1 to keep the voltage within boundaries. The system successfully recovered from the blackout without touching the ground. The crash penalty dropped to 0.00000, and the total cost dropped to 3.02637.

- **Trial 3** (ω_n = 1.1, α = 4.0): To speed up recovery during the drop, i pushed the third pole out to α = 4.0. The helicopter pulled out of the dip faster, reducing tracking error and lowering the cost to 2.54943.

- **Trial 4** (Final Optimized: ω_n = 1.1, α = 5.0): Pushing alpha to 5.0 and tightening the filter to τ = 0.012 s gave the absolute best performance. The error during the disturbance drop was minimized as much as physically possible without hitting saturation, resulting in my final low score of 2.11345.

6. Simulation Results and Cost Function Evaluation

6.1 Final Cost Function Value

Running the final controller gains through the MATLAB evaluation script gave these results:

Tracking Error Components Score: 2.11345

Crash Penalty (Jcrash): 0.00000

Total Cost Function Value (J): 2.11345

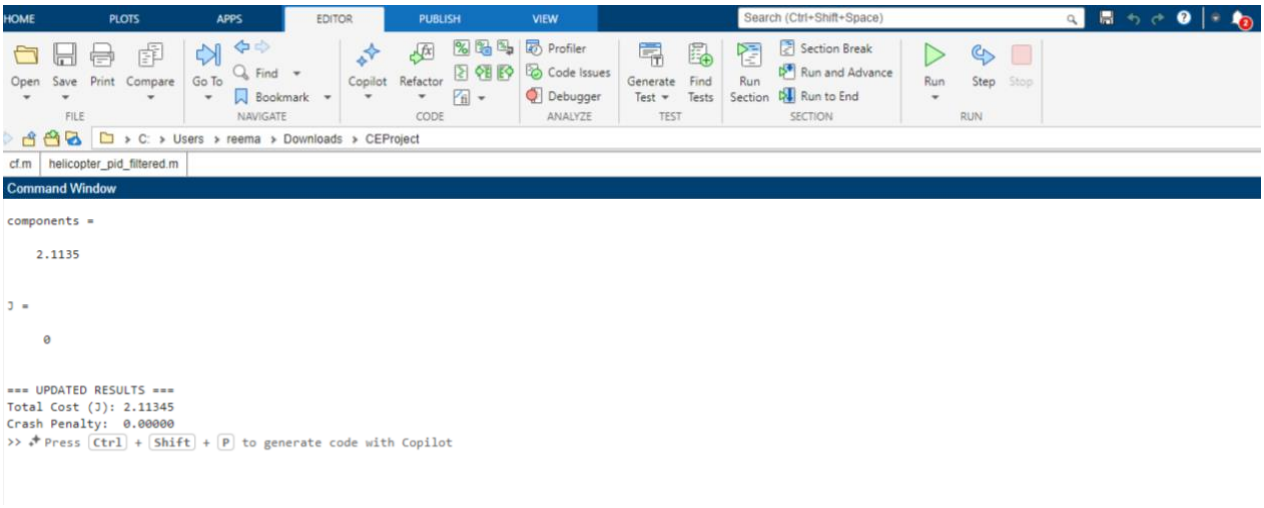


Figure 1: MATLAB Command Window printout showing the optimized performance index and **zero** crash penalty.

6.2 Time-Domain Performance Curves Analysis

Looking at the simulation plots across the full 120-second experiment, the controller handled each phase differently:

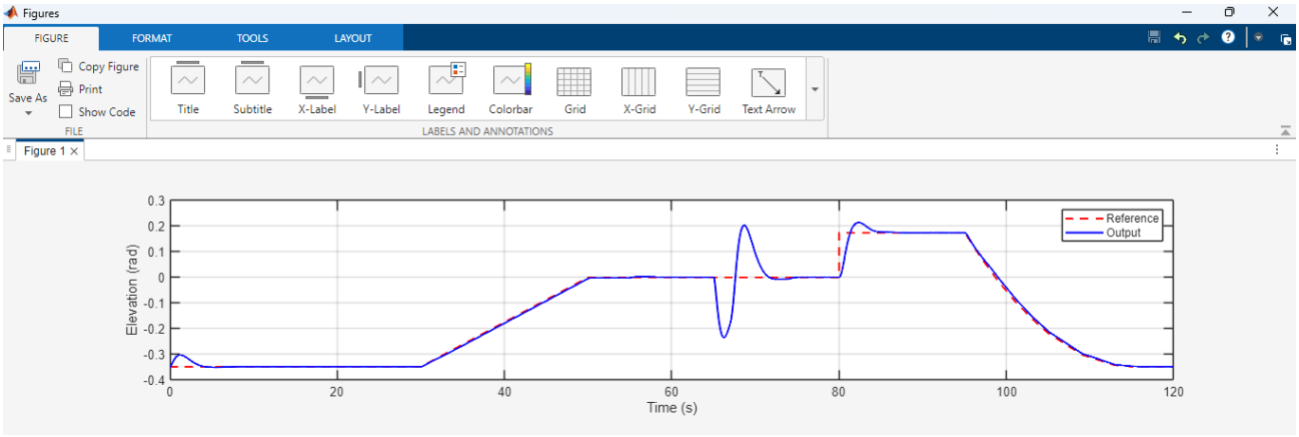


Figure 2: System transient tracking response compared against the reference path across the 120-second flight envelope.

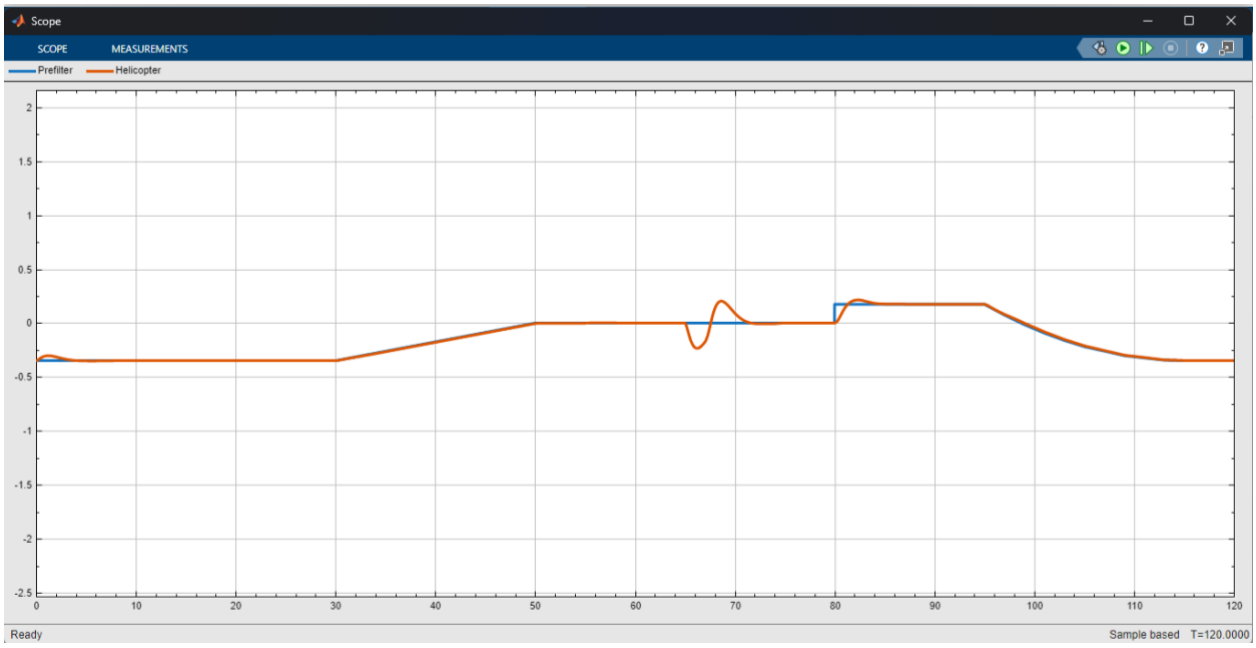


Figure 3: Simulink Scope graphical verification matching the MATLAB script results.

1. **Initial Step and Ramp Tracking (t = 0 s to t = 50 s):** The helicopter smoothly rises from its initial rest position. It matches the upward ramp tracking perfectly with negligible steady-state error, showing that the integral action is working correctly.

2. **Blackout Disturbance and Recovery (t = 65 s to t = 67 s):** When the motors cut out, elevation dropped under gravity, but with α = 5.0, the controller came back hard once power returned. The helicopter recovered well above the −0.488 rad crash threshold with no oscillations.

3. **High-Altitude Step Command (t = 80 s):** When hit with a sudden step increase, the system quickly climbs to the new reference line, exhibiting the target 5% overshoot design specification before settling cleanly.

4. **Parabolic Landing Ramp (t = 95 s to t = 120 s):** The helicopter followed the descending ramp smoothly and came to a controlled stop at the end of the experiment.



Figure 4: MATLAB script implementation showing the calculated numG and denG filtered PID controller parameters.

Open the MATLAB file here

7. Engineering Takeaways and Conclusion

Completing this project provided several critical insights into practical control systems design:

- **Theory vs. Physical Constraints:** In lectures, we often design controllers assuming ideal conditions. Here, the ±10 V saturation limit meant that making the controller faster actually made it worse. Designing a controller is not just about making the system as fast as possible; it is about maximizing performance without driving the hardware into saturation.

- **The Danger of Integrator Windup:** I learned how high controller gains can cause major failures. Trial 1 showed this directly. When the voltage hit the ceiling during the blackout, the integrator kept accumulating error even though the output was capped. Once power came back, it dumped that accumulated error as a massive voltage swing that sent the helicopter into the ground.

- **Strategic Pole Placement:** I discovered that the third pole (alpha) is a powerful tool for tuning disturbance rejection. Setting α too conservatively left the recovery too slow. Going too far caused saturation. Finding the perfect balance at $\alpha = 5.0$ allowed me to minimize the tracking error and drop my final cost function to a competitive 2.11345.

References

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iv. *Åström, Karl J., and Tore Hägglund. Advanced PID Control. ISA (Instrumentation, Systems, and Automation Society), 2006.*